DATA ANALYTICS PORTFOLIO

SALES & PROFITABILITY ANALYSIS

A retail performance case study using Sample Superstore data

PROJECT SNAPSHOT	DETAIL
Prepared by	Darshana Narjari
Dataset	Sample Superstore
Analysis period	2014–2017
Tools	Microsoft Excel • Power Query • PostgreSQL • Power BI
Deliverable	Three-page interactive Power BI dashboard and analytical report

PROJECT FOCUS Understand where sales are generated, which products and categories create profit or loss, and how discounting and geography relate to profitability.

BUSINESS PERFORMANCE • PROFITABILITY • DATA STORYTELLING

01 | AT A GLANCE

Executive Summary

This project examines retail sales and profitability using 9,994 transaction-level records from the Sample Superstore dataset. The analysis covers orders placed between 3 January 2014 and 30 December 2017 and evaluates performance across time, product categories, sub-categories, regions, customer segments, shipping modes, and discount levels.

On the cleaned analytical dataset, the business generated $2.30 million in sales and $286.40 thousand in profit, with an overall profit margin of 12.47%. The results show that overall profitability was positive, but performance varied considerably across the business.

KEY PERFORMANCE INDICATOR	RESULT
Total sales	$2,297,200.65
Total profit	$286,396.59
Unique orders	5,009
Units sold	37,873
Overall profit margin	12.47%
Sales records	9,994

What the analysis shows

- Technology was the strongest main category, leading both sales ($836,154.02) and profit ($145,455.54).
- Furniture generated $741,999.74 in sales but only $18,451.10 in profit, a 2.49% margin.
- Tables recorded the largest sub-category loss at $17,725.58; Bookcases and Supplies also recorded losses.
- The West region led sales and profit, while the Consumer segment contributed the largest absolute sales and profit totals.
- Discount levels of 30% and above were associated with negative aggregate profit margins in this dataset.

INTERPRETATION Sales volume alone does not explain business performance. Profit, margin, product mix, and discount levels should be reviewed together.

02 | WHY THIS ANALYSIS

Business Context, Problem & Objectives

Business context

Retail performance is often assessed through revenue, but revenue by itself can hide weak margins and loss-making products. A product may sell frequently or generate high sales value while contributing little profit after discounts and other costs reflected in the dataset.

Problem statement

The central business question is: where is the business generating sales and profit, and where is profitability under pressure? The project uses historical transaction data to identify patterns that warrant further investigation.

Project objectives

1. Measure overall sales, profit, order volume, quantity, and profit margin.
2. Compare annual and monthly sales and profit trends.
3. Identify the strongest and weakest categories, sub-categories, and products.
4. Compare performance across regions, customer segments, and shipping modes.
5. Examine how recorded discount levels are associated with aggregate profitability.
6. Translate the analysis into practical, evidence-based recommendations.

Key analytical questions

- How did sales and profit change from 2014 through 2017?
- Which product groups generated the most profit and the largest losses?
- Which regions and customer segments contributed most to business results?
- At which discount levels did aggregate profit become negative?

SCOPE This is a descriptive analysis of the fields available in the Sample Superstore dataset. It does not establish the causes of profit differences.

03 | SOURCE AND QUALITY

Dataset Overview & Data Preparation

Dataset profile

ATTRIBUTE	DETAIL
Source	Sample Superstore retail transaction dataset supplied for this project
Records	9,994 transaction rows
Original fields	21 columns
Analytical fields	22 columns, including derived Profit Margin
Order date range	3 Jan 2014 – 30 Dec 2017
Ship date range	7 Jan 2014 – 5 Jan 2018
Currency	U.S. dollars (USD)
Order volume	5,009 unique Order IDs

Preparation and quality checks

- Reviewed column names, data types, date fields, and numeric fields.
- Checked for missing values and exact duplicate rows; none were identified in the supplied raw and cleaned files.
- Checked Row ID uniqueness; no duplicate Row IDs were found in the cleaned dataset.
- Retained negative profit values because they represent loss-making transactions, not invalid records.
- Added a row-level Profit Margin field, calculated as Profit divided by Sales, with a zero-sales safeguard.
- Standardized Sales and Profit to two decimal places in the cleaned analytical file.

Precision and reconciliation note

The raw file stores some Sales and Profit values with more than two decimal places. The cleaned file and SQL table definition use two-decimal monetary precision. Raw full-precision totals are $2,297,200.8603 in Sales and $286,397.0217 in Profit; the cleaned analytical totals are $2,297,200.65 and $286,396.59. The differences are approximately $0.21 and $0.43 respectively. All report KPIs and segment comparisons in this document use the cleaned dataset consistently.

DATA QUALITY RESULT 9,994 rows • 0 missing cells in the supplied cleaned file • 0 exact duplicate rows • 0 duplicate Row IDs.

04 | HOW THE WORK WAS DONE

Methodology, Tools & KPI Definitions

Analysis workflow

STAGE	ACTIVITY	OUTPUT
1. Inspect	Review raw fields, types, dates, missing values, duplicates and monetary precision.	Data profile and quality checks
2. Prepare	Standardize data types and monetary precision; derive Profit Margin.	Cleaned 22-column dataset
3. Query	Use PostgreSQL aggregations for KPIs and breakdowns by time, product, region, segment and discount.	Reusable SQL analysis
4. Visualize	Build KPI cards, trend charts, comparisons, slicers and profitability visuals in Power BI.	Three-page interactive dashboard
5. Interpret	Compare sales with profit and margin; identify areas requiring review.	Findings and recommendations

Tools used

- Microsoft Excel - data inspection, validation, and tabular review.
- Power Query - data cleaning and transformation.
- PostgreSQL - structured queries, aggregation, and analytical calculations.
- Power BI - interactive dashboard, KPI cards, filters, and visual comparisons.

Core KPI definitions

METRIC	DEFINITION
Total Sales	Sum of Sales across all transaction records.
Total Profit	Sum of Profit across all transaction records.
Unique Orders	Count of distinct Order IDs.
Quantity Sold	Sum of Quantity.
Profit Margin	Total Profit ÷ Total Sales × 100. This is a weighted overall margin.
Category / group margin	Group-level total Profit ÷ group-level total Sales × 100.

IMPORTANT For aggregate comparisons, profit margin is calculated from summed profit divided by summed sales. This avoids treating every transaction as equally weighted.

05 | PERFORMANCE OVER TIME

Overall Performance & Time Trends

Overall business performance

The cleaned dataset contains 9,994 sales records across 5,009 distinct orders. Total sales were $2,297,200.65 and total profit was $286,396.59. The resulting weighted profit margin was 12.47%. The business therefore generated positive aggregate profit over the four-year period, although this overall result masks substantial differences between product groups and discount levels.

Annual performance

YEAR	SALES	PROFIT	SALES CHANGE VS PRIOR YEAR
2014	$484,247.47	$49,543.85	—
2015	$470,532.39	$61,618.45	-2.83%
2016	$609,205.74	$81,794.93	+29.47%
2017	$733,215.05	$93,439.36	+20.36%

Sales declined slightly in 2015 before rising in 2016 and 2017. Profit increased in each year, from $49.5 thousand in 2014 to $93.4 thousand in 2017. The strongest annual sales and profit totals occurred in 2017.

How to read the trend

- 2015 shows that sales and profit do not always move in the same direction: sales declined while profit increased.
- 2016 and 2017 recorded stronger sales growth, with profit also increasing.

06 | PRODUCT MIX

Category & Sub-Category Profitability

Main category performance

CATEGORY	SALES	PROFIT	PROFIT MARGIN
Technology	$836,154.02	$145,455.54	17.40%
Office Supplies	$719,046.89	$122,489.95	17.04%
Furniture	$741,999.74	$18,451.10	2.49%

Technology led both sales and profit and delivered the highest category margin. Office Supplies generated slightly less sales than Furniture but substantially more profit. Furniture's low 2.49% margin indicates that its high sales total translated into comparatively little profit.

Sub-categories requiring attention

SUB-CATEGORY	SALES	PROFIT	MARGIN
Tables	$206,965.54	-$17,725.58	-8.56%
Bookcases	$114,879.98	-$3,472.59	-3.02%
Supplies	$46,673.52	-$1,189.08	-2.55%
Copiers	$149,528.01	$55,617.88	37.20%
Phones	$330,007.10	$44,516.16	13.49%
Accessories	$167,380.31	$41,936.78	25.05%

Copiers generated the highest sub-category profit and a strong margin. Tables produced the largest sub-category loss. These results support reviewing product mix, pricing, discounts, and cost structure within Furniture.

KEY TAKEAWAY Compare category sales and profit together. Furniture's sales total was close to Technology's, but its profit was far lower.

07 | WHERE PERFORMANCE OCCURS

Regional, Segment & Shipping Performance

Regional performance

REGION	SALES	PROFIT	MARGIN
West	$725,457.75	$108,418.30	14.94%
East	$678,781.30	$91,522.55	13.48%
South	$391,721.83	$46,749.49	11.93%
Central	$501,239.77	$39,706.25	7.92%

The West region generated the highest sales and profit and also had the strongest regional margin. Central recorded the lowest regional profit margin at 7.92%, despite sales exceeding $500 thousand. This makes Central a useful priority for further investigation.

Customer segment performance

SEGMENT	SALES	PROFIT	MARGIN
Consumer	$1,161,401.11	$134,118.69	11.55%
Corporate	$706,146.33	$91,979.13	13.03%
Home Office	$429,653.21	$60,298.77	14.03%

Consumer generated the highest total sales and profit, while Home Office achieved the highest profit margin of 14.03%, indicating stronger profitability relative to its sales

Shipping mode

SHIP MODE	SALES	PROFIT	UNIQUE ORDERS
Standard Class	$1,358,215.84	$164,088.79	2,994
Second Class	$459,193.40	$57,446.29	964
First Class	$351,428.32	$48,969.73	787
Same Day	$128,363.09	$15,891.78	264

Standard Class represented the largest share of sales and orders. These figures describe the distribution of activity; they do not establish that one shipping mode is more cost-efficient because shipping costs are not included.

08 | DISCOUNT LEVELS

Discount & Profitability Analysis

Discounts can support demand and conversion, but high discounts may reduce the profit retained from each sale. This section compares total sales, total profit, and weighted profit margin at each discount level in the cleaned dataset.

DISCOUNT	SALES	PROFIT	WEIGHTED MARGIN
0%	$1,087,908.47	$320,987.06	29.50%
10%	$54,369.30	$9,029.21	16.61%
15%	$27,558.53	$1,418.98	5.15%
20%	$764,594.28	$90,337.62	11.82%
30%	$103,226.68	-$10,369.32	-10.05%
32%	$14,493.45	-$2,391.16	-16.50%
40%	$116,417.83	-$23,057.05	-19.81%
45%	$5,484.98	-$2,493.12	-45.45%
50%	$58,918.53	-$20,506.50	-34.80%
60%	$6,644.68	-$5,944.62	-89.46%
70%	$40,620.24	-$40,075.40	-98.66%
80%	$16,963.68	-$30,539.11	-180.03%

Interpretation

- At 0% and 20% discount, aggregate profit remained positive.
- At every listed discount level from 30% through 80%, aggregate profit was negative.
- The most extreme negative weighted margin occurred at the 80% discount level (-180.03%).
- Discount groups have different sales volumes and transaction mixes; the table should not be interpreted as a controlled experiment.

CAUTION: ASSOCIATION IS NOT CAUSATION The analysis shows an observed relationship in historical records. It does not prove that discounts alone caused losses; product mix, costs, order characteristics, and other factors may also influence profit.

09 | BEST AND WEAKEST PRODUCTS

Product-Level Profitability

Highest-profit products

PRODUCT	SALES	PROFIT
Canon imageCLASS 2200 Advanced Copier	$61,599.83	$25,199.94
Fellowes PB500 Electric Punch Plastic Comb Binding Machine	$27,453.38	$7,753.04
Hewlett Packard LaserJet 3310 Copier	$18,839.68	$6,983.89
Canon PC1060 Personal Laser Copier	$11,619.83	$4,570.94
HP Designjet T520 Inkjet Large Format Printer - 24-inch Color	$18,374.89	$4,094.98

Largest-loss products

PRODUCT	SALES	PROFIT
Cubify CubeX 3D Printer Double Head Print	$11,099.95	-$8,879.97
Lexmark MX611dhe Monochrome Laser Printer	$16,829.89	-$4,589.97
Cubify CubeX 3D Printer Triple Head Print	$7,999.98	-$3,839.99
Chromcraft Bull-Nose Wood Oval Conference Table	$9,917.64	-$2,876.11
Bush Advantage Collection Racetrack Conference Table	$9,544.72	-$1,934.40

Business interpretation

The Canon imageCLASS 2200 Advanced Copier was the most profitable individual product, contributing $25,199.94 in profit. The Cubify CubeX 3D Printer Double Head Print recorded the largest product-level loss at $8,879.97.

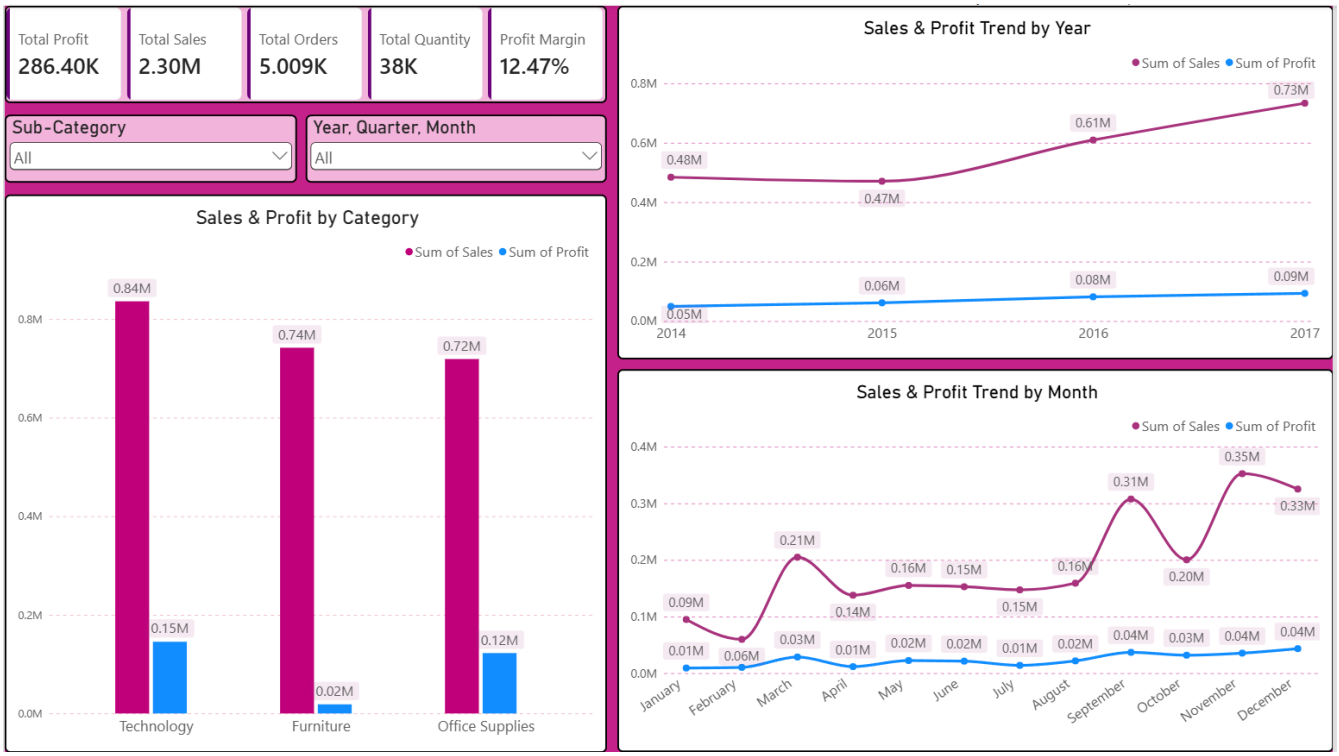
Product-level results can guide deeper investigation, but the dataset does not provide detailed procurement cost, inventory, or competitor-pricing information. The analysis therefore identifies where to investigate rather than prescribing an automatic discontinue or promotion decision.

10 | DASHBOARD PAGE 1

Power BI Dashboard - Sales Overview

The Sales Overview page is designed to provide a fast view of business performance. KPI cards summarize the main measures, while trend and category visuals help users explore how sales and profit are distributed.

Figure 1. Sales Overview dashboard



What this page communicates

- Total Sales, Total Profit, Unique Orders, Quantity Sold, and Overall Profit Margin.
- Annual sales and profit trends across 2014–2017.
- Monthly sales pattern, clearly labelled as a combined month-of-year view if years are aggregated.
- Sales and profit comparison across Furniture, Office Supplies, and Technology.
- Year and category filters for interactive exploration.

READER TIP Start with the KPI cards, then compare the annual trend and category breakdown to understand the overall business picture.

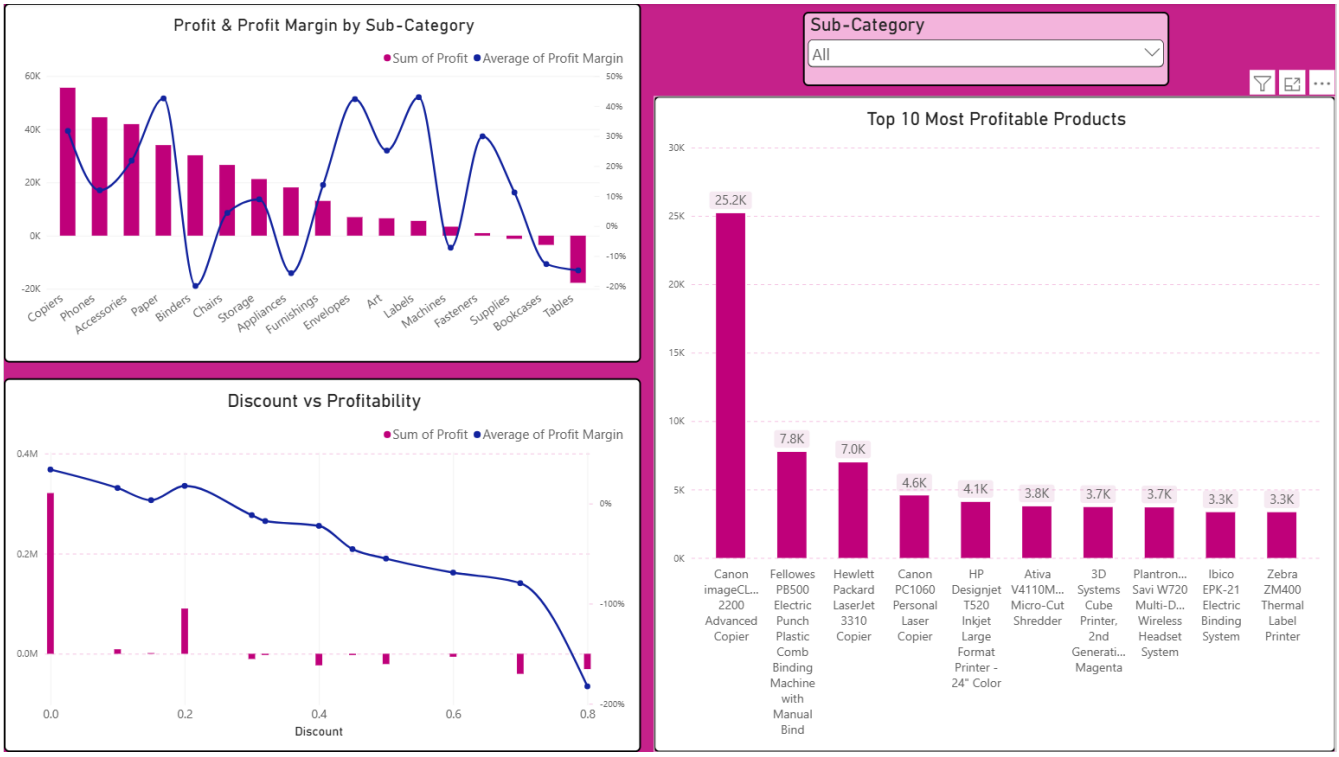
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Power BI Dashboard - Profitability Analysis

The Profitability Analysis page focuses on where the business earns or loses profit. It combines sub-category performance, discount analysis, and product-level results to highlight areas requiring attention.

Recommended screenshot

Figure 2. Profitability Analysis dashboard



What this page communicates

- Profit and weighted profit margin by sub-category.
- Discount level compared with total profit and weighted profit margin.
- Top 10 products by total profit.
- Sub-category filters for more focused analysis.

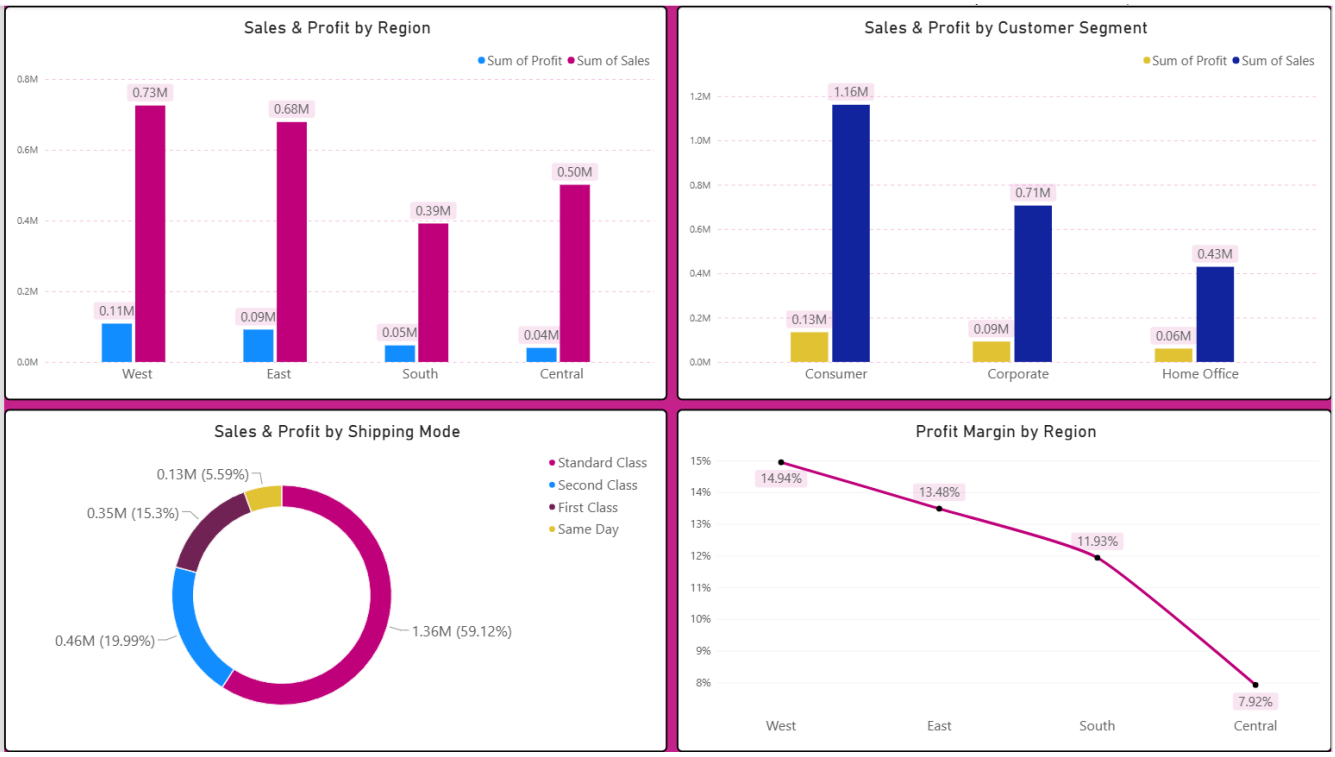
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Power BI Dashboard - Business Performance

The Business Performance page compares results across regions and customer segments and shows how sales are distributed by shipping mode. These views help identify differences in contribution and margin across the business.

Recommended screenshot

Figure 3. Business Performance dashboard



What this page communicates

- Sales and profit by region.
- Sales and profit by customer segment.
- Sales distribution by shipping mode.
- Profit margin comparison across regions.

INTERPRETATION The West led regional sales and profit, while Home Office recorded the highest customer-segment margin. Standard Class accounted for the largest order and sales volume.

13 | FROM ANALYSIS TO ACTION

Key Findings, Recommendations & Conclusion

Key findings

- Technology was the strongest category by sales, profit, and category margin.
- Furniture generated high sales but a low 2.49% margin; Tables and Bookcases were loss-making.
- The West region led regional sales and profit, while Central had the lowest regional margin.
- Consumer contributed the largest absolute sales and profit; Home Office had the highest segment margin.
- Discount levels of 30% and above were associated with negative aggregate profit in the observed data.
- Product profitability varied substantially, from $25,199.94 profit for the leading product to an $8,879.97 loss for the largest-loss product.

Recommendations

PRIORITY	RECOMMENDED ACTION
1. Discount governance	Review high-discount transactions, especially 30% and above. Evaluate margin impact alongside sales volume and product mix.
2. Loss-making products	Investigate the largest-loss products and review price, discount, cost, and demand before deciding on corrective action.
3. Furniture margin	Review pricing, discounting, and product mix in Furniture, with particular attention to Tables and Bookcases.
4. Regional review	Investigate Central's lower margin and compare its product and discount mix with stronger regions.
5. Profit-led planning	Track profit and margin alongside sales when prioritizing products, promotions, and inventory decisions.

Conclusion

The analysis of 9,994 Sample Superstore records shows a business with $2.30 million in sales and $286.40 thousand in profit over 2014–2017. Technology and Copiers were strong contributors, while Furniture and selected sub-categories require closer review. Higher discount levels were associated with negative aggregate profitability, and performance varied by region, segment, and product.

The three-page Power BI dashboard presents these patterns in an interactive format. The project demonstrates a complete analytical workflow - from data preparation and PostgreSQL querying to dashboard communication and business recommendations.

Limitations & next steps

The dataset does not include detailed procurement costs, inventory, marketing spend, competitor prices, or customer lifetime value. The data is historical and the discount analysis is observational. Future analysis can be improved by including product costs and inventory information, studying repeat customers, and testing different promotional offers to identify which strategies are more profitable

Data basis: supplied raw and cleaned Sample Superstore files; PostgreSQL analysis file; Power BI dashboard supplied for this project.